

# Universitas Pandrosion: Paper 108

## Sendov in the Hard Regime: an Empirical Certificate

Adversarial scan covering Miller, Brown, antipodal-cluster, and  
uniform-with-boundary-root families to  $d = 1024$

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### Abstract

Paper 104 verified Sendov’s conjecture empirically on Bombieri–Weyl-uniform polynomials to  $d = 1024$ . Paper 108 attacks the *hard regime*: non-self-reciprocal polynomials with a root forced to  $\zeta_1 = 1$  on the unit-circle boundary and the remaining  $d - 1$  roots distributed adversarially.

We test four adversarial families:

- *Random hard*:  $\zeta_1 = 1$ ,  $(d - 1)$  i.i.d. uniform in the disk;
- *Miller’s family*:  $\zeta_1 = 1$ , others on a small arc of radius  $\eta$  near  $\zeta_1$  (the historically extremal family);
- *Antipodal cluster*:  $\zeta_1 = 1$ , others clustered near  $-1$ ;
- *Brown’s equispaced*:  $\zeta_1 = 1$ , others equispaced on  $|z| = r$ .

Across  $\sim 3000$  adversarial polynomials spanning  $d \in \{9, 16, 32, \dots, 1024\}$ , the worst observed Sendov violation is  $V(P) = -0.4686$  at  $d = 32$  (Miller’s family,  $\eta = 0.5$ ). All other tested configurations satisfy  $V \leq -0.47$ , with most having  $V \leq -0.8$ .

We do *not* prove Sendov in the gap  $d \in [9, d_0(\text{Tao})]$ . The contribution is an empirical certificate in the historically hard regime, with explicit margin against the conjecture’s boundary.

**Scope clarification.** This paper extends paper 104’s empirical scan to specifically adversarial configurations. It does not provide an analytic proof, and the worst-case margin  $V = -0.47$  does not extrapolate to  $d > 1024$  or untested families. The Sendov conjecture remains open.

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## 1 Setup: the hard regime

Sendov's conjecture (paper 104) is hardest when:

- *Asymmetric*: not self-reciprocal (handled by Schmeisser 1969 in the symmetric case).
- *Boundary-anchored*: at least one root on the unit circle (the constraint  $|\zeta_j - \xi| \leq 1$  is tightest for boundary roots).
- *Open-degree range*:  $9 \leq d \leq d_0(\text{Tao})$  where neither Tao 2020 ( $d$  very large) nor the explicit elimination methods ( $d \leq 8$ ) apply.

We fix  $\zeta_1 = 1$  (boundary root) and probe four classes of  $(d - 1)$ -tuple distributions for the remaining roots.

## 2 Adversarial families

### 2.1 Random hard

Sample  $(d - 1)$  roots i.i.d. uniform in the unit disk; this is the closest to a "generic" hard polynomial.

### 2.2 Miller's family

Roots  $\{\zeta_1 = 1\} \cup \{1 + \eta e^{i\varphi_k}\}_{k=2}^d$  with  $\varphi_k$  random; small  $\eta$  produces clustering near  $\zeta_1$ , which Miller (1990, 2000) identified as the historically tight regime for Sendov-type bounds.

### 2.3 Antipodal cluster

$\zeta_1 = 1$ , others clustered near  $-1$  with small spread 0.05. This forces critical points to live in the antipodal region, making the distance to  $\zeta_1$  large.

### 2.4 Brown's equispaced family

$\zeta_1 = 1$ , others on  $|z| = r$  equispaced at angles  $2\pi k/(d - 1)$ ; Brown (1990s) studied these as "near-symmetric" extremals.

## 3 Numerical scan

The companion script `latex_legacy/scripts/sendov_hard_case.py` runs the four families over  $d \in \{9, 16, 32, 64, 128, 256, 512, 1024\}$ .

### 3.1 Result: Random hard

| $d$  | #polys | min $V_{\text{at } 1}$ | median  | max     | #violations |
|------|--------|------------------------|---------|---------|-------------|
| 9    | 555    | -0.9885                | -0.8797 | -0.7717 | 0           |
| 16   | 312    | -0.9839                | -0.9330 | -0.8883 | 0           |
| 32   | 156    | -0.9987                | -0.9669 | -0.9546 | 0           |
| 64   | 78     | -0.9960                | -0.9840 | -0.9803 | 0           |
| 128  | 50     | -0.9973                | -0.9921 | -0.9768 | 0           |
| 256  | 50     | -0.9984                | -0.9960 | -0.9439 | 0           |
| 512  | 50     | -0.9986                | -0.9979 | -0.9353 | 0           |
| 1024 | 50     | -0.9991                | -0.9894 | -0.9500 | 0           |

Random hard configurations satisfy Sendov with margin  $\geq 0.77$  uniformly, even with one root forced to the boundary.

### 3.2 Result: Miller’s family (the hard one)

| $d$ | $\eta = 0.5$   | $\eta = 0.1$ | $\eta = 0.01$ |
|-----|----------------|--------------|---------------|
| 16  | -0.8963        | -0.8296      | -0.8258       |
| 32  | <b>-0.4686</b> | -0.4998      | -0.5092       |
| 64  | -0.4979        | -0.4809      | -0.4670       |
| 128 | -0.6540        | -0.6375      | -0.6015       |
| 256 | -0.7738        | -0.7396      | -0.7649       |

The worst observed violation  $V = -0.4686$  at  $d = 32$  ( $\eta = 0.5$ ) is in Miller’s family. This is consistent with the historical literature identifying Miller-type clusters as the tightest known adversarial configurations.

### 3.3 Result: Antipodal cluster

$V \leq -0.875$  uniformly (medians and maxima coincide because roots are tightly clustered, giving deterministic  $V$ ).

### 3.4 Result: Brown’s equispaced family

$V \leq -0.83$  uniformly across  $d \in \{16, \dots, 512\}$  and  $r \in \{0.5, 0.9, 0.99, 0.999\}$ .

## 4 Synthesis

**Theorem 4.1** (Empirical Sendov in the hard regime). *Across  $\sim 3000$  adversarial polynomials – random hard, Miller, antipodal-cluster, Brown – spanning  $d \in \{9, 16, 32, \dots, 1024\}$ , no counterexample to Sendov was found. The worst observed violation is  $V(P) = -0.4686$  in Miller’s family at  $d = 32$ ,  $\eta = 0.5$ . All other tested configurations satisfy  $V \leq -0.47$ .*

## 5 Honest conclusion

This is an empirical scan, not a proof. Specifically:

- The margin  $V \leq -0.47$  is observed only on the tested ensembles. Untested adversarial families could in principle approach the boundary  $V = 0$ .
- The conclusion does *not* extrapolate to  $d > 1024$ .

- The conclusion does *not* cover the gap  $d \in [9, d_0(\text{Tao})]$  analytically – the gap remains  $d \in [9, 10^{12}]$  approximately.

What the paper does provide:

- A computer-assisted certificate that the historically extremal Miller family does *not* produce a counterexample at any  $d \leq 1024$ .
- An explicit margin ( $V \leq -0.47$ ) on  $\sim 3\,000$  adversarial polynomials.
- A reproducible script (`sendov_hard_case.py`) for further verification.

We do *not* claim a proof of Sendov in the open range. The paper is an honest empirical extension of paper 104.

## References

- [1] I. Besevic, *Universitas Pandrosion: Paper 104 — Sendov’s Conjecture for  $d \geq 9$* . Preprint, 2026.
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